Sheet 29 Mon 11 Jan. 2027 *Grouped data: histogram and cumulative frequencies*

1. Warm-up · Quick Give the width and the midpoint of the class $[20;35[$.
2. Core · Table Classes $[0;10[$, $[10;20[$, $[20;30[$ with frequencies (counts) 4, 6, 10. Give the relative frequencies, then the cumulative increasing relative frequencies.
3. Core + · Quick In a histogram with classes of equal width, two rectangles have heights 3 cm and 6 cm. What is the ratio of their frequencies (counts)?
4. Review · Quick Write 0.00032 in scientific notation.
5. Thread · Quick Solve in $\mathbb{R}$: $2(x+3) = 5x - 9$.

Sheet 30 Tue 12 Jan. 2027 *Weighted mean, median class; spreadsheet and Python*

1. Warm-up · Quick Calculate $\dfrac{2\times 8 + 3\times 12}{5}$.
2. Core · Method Classes $[0;10[$, $[10;20[$, $[20;30[$ with frequencies (counts) 4 ; 7 ; 9. Estimate the mean using the class midpoints; give the median class.
3. Core + · Code reading What does this programme calculate and what value does it give?

```
notes = [12, 15, 9, 14]
s = 0
for n in notes:
    s = s + n
print(s / len(notes))
```

4. Review · Quick An item costing 45 000 Ar is sold at $-30\,\%$. What is the sale price?
5. Thread · Prove Programme: choose n; add 1; square it; subtract n^2. Show that the result is always an odd number.

Algebra, equations and inequalities

Sheet 31 Thu 14 Jan. 2027 *Integer powers (positive and negative)*

1. Warm-up · Quick Calculate 2^{-3}, 10^{-2} and $(-3)^2$.
2. Core · Method Simplify $\dfrac{2^3 \times 2^{-5}}{2^{-1}}$.
3. Core + · Method For $x \neq 0$, write $(x^2)^{-3} \times x^7$ in the form x^n.
4. Review · Quick Median of the series 4, 7, 9, 10, 12, 15?
5. Thread · Identify Which solid has a disc-shaped base, a curved lateral surface and a single vertex?

Sheet 32 Mon 18 Jan. 2027 *Square roots; $\sqrt{a^2} = |a|$*

1. Warm-up · Quick Calculate $\sqrt{36}$ and $\sqrt{0.09}$.
2. Core · Method Write $\sqrt{72}$ in the form $a\sqrt{2}$ and calculate $\sqrt{(-5)^2}$.
3. Core + · True/false True or false? Justify. $\sqrt{9+16} = \sqrt{9} + \sqrt{16}$.
4. Review · Quick Calculate $(3^2)^3$.
5. Thread · True/false True or false? Justify. The diagonals of a rhombus are perpendicular.

Sheet 33 Tue 19 Jan. 2027 *Fractional expressions; expanded, reduced or factorised forms*

1. Warm-up · Quick Calculate $\dfrac{1}{2} + \dfrac{1}{3}$.
2. Core · Method Write as a single fraction $\dfrac{1}{x} + \dfrac{1}{x+1}$ (for $x \neq 0$ and $x \neq -1$).
3. Core + · Method Factorise $x^2 - 4$ then simplify $\dfrac{x^2-4}{x-2}$ for $x \neq 2$.
4. Review · Quick Calculate $2a - 3b$ for $a = 2$ and $b = -1$.

5. Thread · Quick $ABCDEFGH$ is a cuboid with $AB = 4$, $AD = 3$, $AE = 2$. In the frame $(A; B, D, E)$, what are the coordinates of G?

Sheet 34 Thu 21 Jan. 2027 *Inequalities: sum, multiplication by a real number*

1. Warm-up · Quick Complete with $<$ or $>$: $-3 \ldots -2$; $\frac{1}{3} \ldots 0.3$.
2. Core · Method We have $a < b$. Compare $-2a$ and $-2b$, then $a + 5$ and $b + 5$.
3. Core + · Method Given $2 < x < 3$ and $1 < y < 4$, find bounds for $x + y$ then $x - y$.
4. Review · Quick Calculate $(-3) \times (-4) + 2 \times (-5)$.
5. Thread · Problem A 120 cm by 84 cm sheet is cut into identical squares, with no waste. What is the largest possible side length?

Sheet 35 Mon 25 Jan. 2027 *Comparing by difference and by quotient*

1. Warm-up · Quick Compare $\frac{2}{3}$ and $\frac{3}{4}$ by studying the sign of the difference.
2. Core · Method Compare $x^2 + 1$ and $2x$ for every real number x.
3. Core + · Method Compare $\frac{7}{5}$ and $\frac{8}{6}$ by quotient (positive numbers).
4. Review · Table Monthly rainfall (fictional data).

Month	Jan.	Feb.	Mar.	Apr.
Rain (mm)	280	300	240	120

Which month is the wettest? What is the total over these 4 months?

5. Thread · Table → formula x: -2, -1, 0, 1 and $f(x)$: 4, 1, 0, 1. Is the function affine? Suggest a formula.

Sheet 36 Tue 26 Jan. 2027 *First-degree equations and inequalities; $x^2 = a$; zero product*

1. Warm-up · Quick Solve $3x - 5 = 7$.
2. Core · Method Solve $2x + 3 \leqslant 7 - x$.
3. Core + · Method Solve $(x - 2)(2x + 6) = 0$ then $x^2 = 7$.
4. Review · Quick Compare 1.4 and $\sqrt{2}$ without a calculator.
5. Thread · Quick ABC is isosceles at A with $\widehat{BAC} = 40°$. What is the measure of $\widehat{ABC}$?

Sheet 37 Thu 28 Jan. 2027 *Quotient equation $\frac{A(x)}{B(x)} = k$*

1. Warm-up · Quick For which value of x is the expression $\frac{1}{x - 3}$ undefined?
2. Core · Method Solve $\frac{2x - 1}{x + 3} = 0$.
3. Core + · Method Solve $\frac{x + 1}{x - 2} = 3$.
4. Review · Expand Expand and simplify $(x + 2)(x - 5)$.
5. Thread · Code reading What do $g(3)$ and $g(-5)$ return?

```
def g(x):
    if x > 0:
        return 2*x
    else:
        return -x
```